Applications of binomial distribution

1

a $\frac{125}{3888}$ b $\frac{625}{648}$ c $\frac{3875}{3888}$

2 0.02

3 0.0686

4 0.0839

5 0.2446

6 0.0002

7

a 0.003318 b 0.0333 c 0.1996

8 0.0705

9 0.8750

10 0.1356

11 5.7×10^{-11}

12

a Binomial, 0.4 is the probability of "success" - in this case, selecting a dinosaur to play with.

b

i 0.161 ii 0.071 iii 0.930

13 70

14

a 7.5 days

b The chance of rain on any given day is not independent of the weather on previous days.

15 $n = 8$

16

a 12 b 9 c 3

17

a 15 b 12

18

a i 8 ii 8 iii 2.53 iv 2.65

b X

c 0.68 d 0.66

19

a 13.72 b 9.9

20

a 720 b 25.17

21

a $n = 25,\ p = 0.2$ b 0.36

22

a 0.2363 b 0.9102 c 0.7338 d 0.9630

e 0.0001

23

a 0.00047 b 0.11806 c 0.02449 d 0.21321

24

a 8 b $\dfrac{4\sqrt{10}}{5}$

c $P\left(3 \leq X \leq 13\right) = 0.97$

25

a 0.85

b 0.15

c $1 - (0.98)^n$

d 149

e $Y \sim B\left(12, 0.15\right)$, hence $P\left(Y = y\right) = \binom{12}{y} \times 0.15^y \times 0.85^{12-y}$

f 0.91

26

a 0.0256 b 1

27

a $p = \dfrac{3}{8}$ b 0.86

c $P(X = x) = \binom{12}{x} \times 0.86^x \times 0.14^{12-x}$ d 0.29

28

a

i 0.0235 ii 0.0775 iii 0.0040

b Yes, the event is unlikely to happen by chance. It is more likely there is a problem in the kitchen.

29

a 0.0197 b 0.0207 c 0.7977 d 0.9870

30

a 0.0001 b 0.0198 c 1.3×10^{-22} d 0.9967

e 0.9997

31

a 0.0006 b 0.0250 c $n = 6$

32

a 0.2 b 0.9895 c 0.2208 d $n = 21$

33

a 0.00047 b 0.0117 c 1.1×10^{-25}

34

a 0.0115 b 0.0069 c 0.0188

35

a $\dfrac{25}{1296}$

b $\dfrac{25}{216}$

c $X \sim B\left(10, \dfrac{25}{216}\right)$, hence $P(X = x) = \binom{10}{x} \times \left(\dfrac{25}{216}\right)^x \times \left(\dfrac{191}{216}\right)^{10-x}$, for $x = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$ and 10.

d 0.90

36

a ${}^{n}C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{n-3}$ for $n = 3, 4, 5, \ldots$

b 10

37

a 0.1664

b 0.2717

c 8.790×10^{-7}

d 12

e 11.7

f 0.01294

g 0.1006